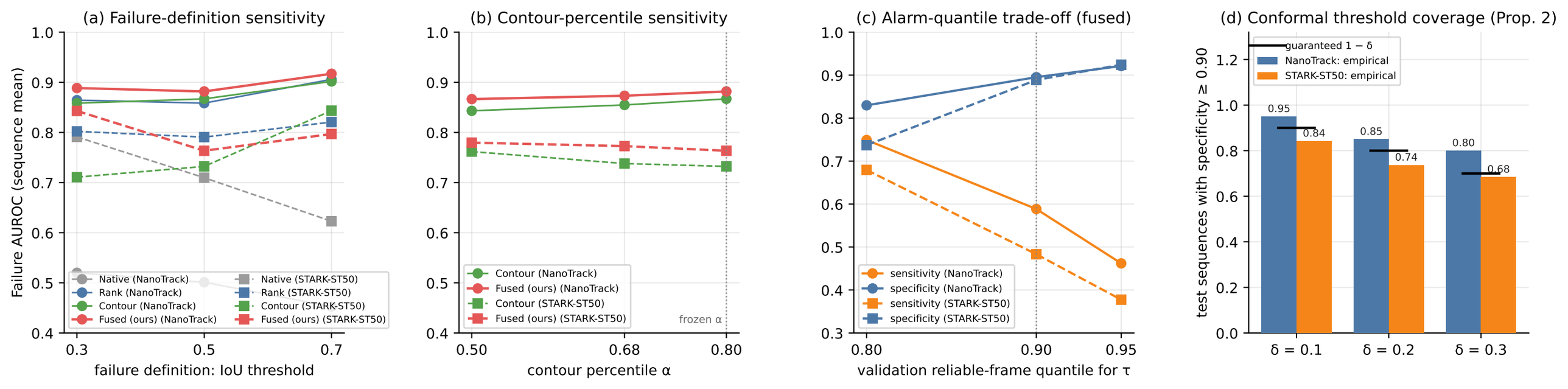
(a) Failure-definition sensitivity
Failure AUROC (sequence mean)
failure definition: IoU threshold
Native (NanoTrack)
Rank (NanoTrack)
Contour (NanoTrack)
Fused (ours) (NanoTrack)
Native (STARK-ST50)
Rank (STARK-ST50)
Contour (STARK-ST50)
Fused (ours) (STARK-ST50)
(b) Contour-percentile sensitivity
contour percentile α
Contour (NanoTrack)
Fused (ours) (NanoTrack)
Contour (STARK-ST50)
Fused (ours) (STARK-ST50)
frozen α
(c) Alarm-quantile trade-off (fused)
validation reliable-frame quantile for τ
sensitivity (NanoTrack)
specificity (NanoTrack)
sensitivity (STARK-ST50)
specificity (STARK-ST50)
(d) Conformal threshold coverage (Prop. 2)
test sequences with specificity ≥ 0.90
guaranteed 1 − δ
NanoTrack: empirical
STARK-ST50: empirical
0.95
0.84
0.85
0.74
0.80
0.68
δ = 0.1
δ = 0.2
δ = 0.3